Guidodynamics Dynamometer

User Guide

Capstone Team F25-66

Modular Vehicle Dynamometer.

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Notice: This document is intended for users of the Guidodynamics Modular Vehicle Dynamometer. For developer and hardware documentation, refer to the DAQ Developer Guide and DAQ Module Datasheet.

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1 Overview

This document contains information relevant to an user of Capstone Team F25-66's Modular Vehicle Dynamometer. Any user seeking to go beyond use of the dynamometer should reference the DAQ Developer Guide, and the DAQ Module Datasheet, linked here.

1.1 Vehicle Compatibility

The dyno is compatible with the following vehicle specifications:

- **Power Output:** <= 100 hp
- **Track Length:** 56-75 in
- **Track Width:** 44-53 in
- **Tire Width:** <= 10 in
- **Top Speed:** 20-90 mph

1.2 Occupational Safety

The dynamometer presents several hazards that all users must understand before, during, and after operation. Anyone present in the same room as the dyno must take appropriate precautions, and the **User Safety** section of this document must be read and understood in full prior to operation.

Failure to comply may result in serious injury or death.

1.3 Software

The Dynamometer comes with comprehensive software to make data collection and visualization easy. The user software stack ingests data from the DAQ module, stores it in a database (InfluxDB), and displays it using Grafana. The software sections of this guide contain step by step instructions for how to configure this system on a Windows 11 computer.

At this time, the system does not virtualize. Keep in mind these steps will install the software on the top level on the computer, potentially interfering with other programs.

2 User Safety

The dynamometer is contained within a specialty built room that shields occupants of the adjacent classroom from most hazards, however, precautions must be taken.

2.1 General Operating Conditions

- The following PPE is required for dyno operation: **safety glasses, closed-toe shoes, long hair tied back, ear protection**
- A minimum of three people must be present during operation, with one stationed by the emergency stop button in the classroom and one stationed by the emergency stop button on the control box.
- The doors between the dynamometer room and the classroom must remain closed while in operation.
- If an ICE powertrain is being tested, the ventilation system must be enabled by turning the switch near the main IBC water tank, and the garage door opened. No spectators shall be allowed behind the dynamometer while in operation.

- For improved ventilation, it is recommended that flexible ducts be connected directly to the exhaust outlet(s). If the gas detection alarm activates during operation, stop the engine and move to fresh air.
- If users must be in the dynamometer room (e.g. in vehicle chassis configuration), they should be clear of the front and rear of the vehicle.

2.2 ESTOP Test

Additionally, before every session, the dynamometers safety system must be tested.

1. Ensure the main power connector is unplugged, and the master power switch it set to off. Unplug both pumps from their L5-20 connectors on the bottom of the panel.
2. Ensure all ESTOP buttons are in the released position. Plug in the mains power connector, and turn the master power switch to on.
3. Press the blue reset button, and press an ESTOP button. Observe the stack light turn red.
4. Repeat steps two and three for each of the two ESTOP buttons.

If the stack light does not turn red or does not light up at all, for any ESTOP, do not use the dynamometer and alert an TA or Professor.

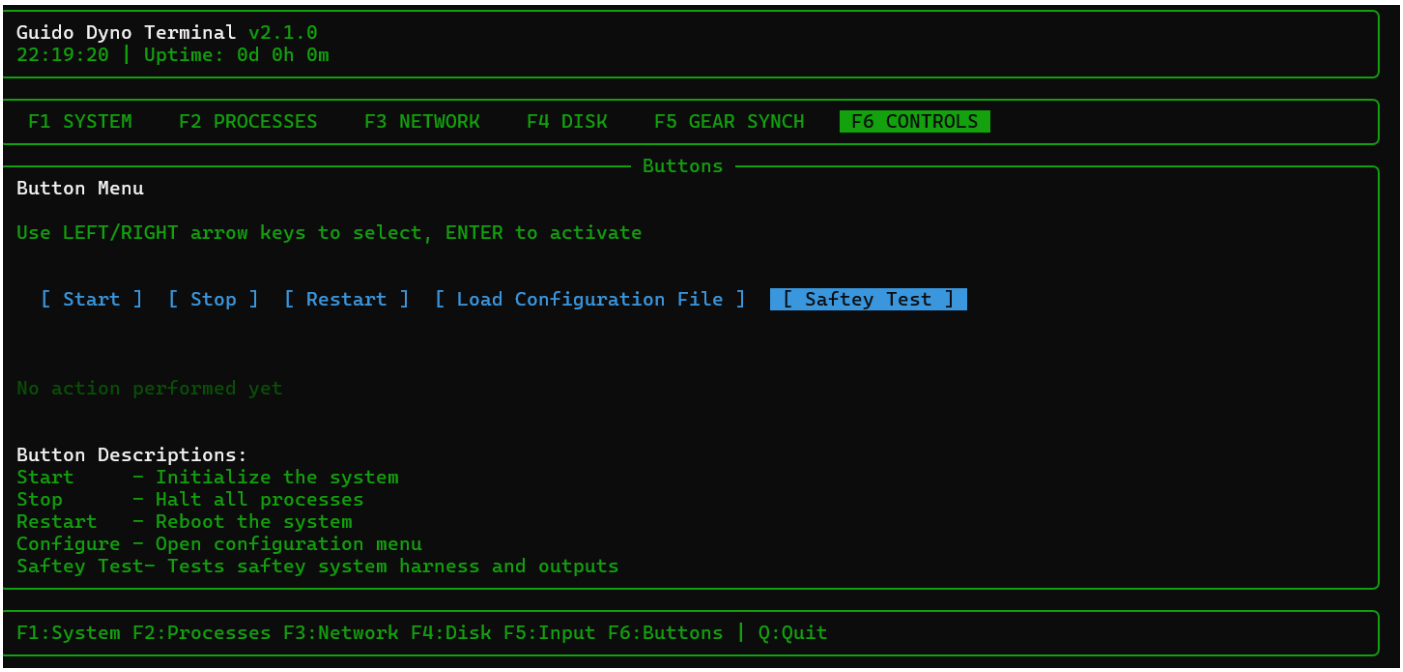
After testing each ESTOP, set the master power switch to off, unplug the dynamometer from the mains, and plug in both pump's L5-20 connectors.

2.3 Software Safety System Test

The software safety system must also be tested before each session.

After testing the ESTOP buttons from the prior section, ensure the dynamometer is powered on, and press the blue reset button to set the stack light green.

To manually perform a safety system test, navigate to panel 6 of the user terminal, and select the button "Safety Test".



Again, the stack light should turn red. If it does not, do not use the dyno and alert an TA or Professor.

2.4 Pre-Operative Checklist

Complete the following checklist before any dynamometer session:

1. Complete the ESTOP Test as prescribed in 2.2
2. Complete the Software Safety System Test as prescribed in 2.3
3. Ensure all conditions in 2.1 are met.
4. Set the **Main IBC Valve** (located at the bottom of the IBC tank) to the OPEN position.
5. Ensure that both ball valves (located at the outlet of the pump and the outlet of the return tank) are set to the OPEN position.
6. Inspect rotating machinery, including user DUTs (devices under test), and ensure fabric, rope, and other loose items are secured.
7. If in vehicle mode, ensure vehicle is properly strapped to the dynamometer and is unable to move. Also ensure that the water brake shaft pointing towards the left tire is covered to avoid tire damage.
    - Straps must connect to rigid frame members. **Do not connect straps to body work, suspension components, or drivetrain components**.
8. If in engine or brakes mode, ensure test stand is secured at all available points.
9. Turn the master power switch to the ON position and press the blue reset button. The stack light should turn green.

2.5 Shut-Off Checklist

Complete the following checklist after a dyno session:

1. Turn the master power switch to the OFF position.
2. Set the **Main IBC Valve** (located at the bottom of the IBC tank) to the CLOSED position.
3. Turn off the ventilation system.
4. Turn off lights and lock all doors to the dyno bay.

3 Software Installation

3.1 System Requirements

- Windows 11
- 8 GB RAM
- Ethernet Network Adapter [1]

3.2 Dependencies

3.2.1 Python

Python 3.13 or later must be installed. Lower versions of python 3 can be used, but are not guaranteed. To check this, open command prompt, and run the following:

```
python --version
```

You should see:

```
Python 3.x.x
```

If you do not see your python version, or you do not have Python 3.13+, install the latest version of python (here) on your computer before proceeding.

Long PATH Names

Depending on your computer, the python installer may ask to adjust path lengths, universal path settings, or install CPython. Respond "y" to these requests.

3.2.2 Github Desktop

Make sure Github desktop is installed on your computer.[2] Default settings are OK. Then sign in to your Github account.

Download the latest version here

3.2.3 Grafana

Download Grafana from this website. Make sure to download it here, instead of the Grafana Cloud website. Select version 12.3.2 Enterprise edition.

[1] Even if your computer does not have a physical Ethernet port, you likely still have an virtual Ethernet network adapter. You will still need an USB to Ethernet adapter dongle

[2] Install the Github desktop application even if you already have Git, unless you are sophisticated enough to set up Git LFS on your own

Keep all default settings while going through the Grafana setup steps. After the installation finishes, restart your computer. [3]

3.3 Repository Setup

The GitHub repository can be downloaded multiple different ways. For use when the program is still under development, it is recommended to use GitHub desktop to install and update the program on your local computer. Alternatively, the program can be downloaded as a .zip file, which is a quicker but harder to update method. See the appropriate subsection for how the program should be downloaded and installed on your personal machine.

Which Option

When in doubt, use the Github Desktop method 3.3.1 (SKIP 3.3.2). It is more stable and easier to update.

3.3.1 GitHub Desktop

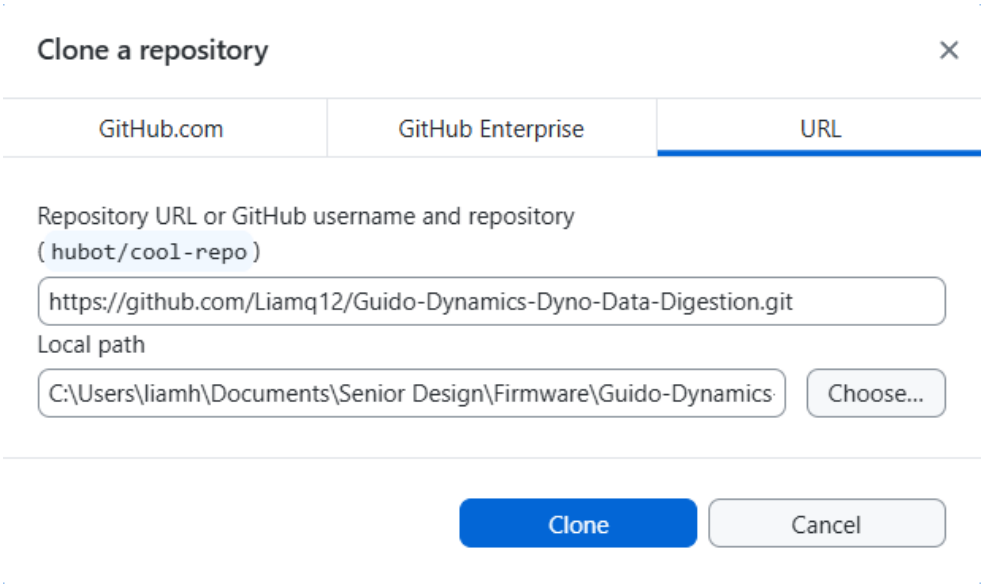
Open Github desktop and navigate:
File → Clone Repository Select the URL menu, and enter the following URL: [4]

https://github.com/Liamq12/Guido-Dynamics-Dyno-Data-Digestion.git

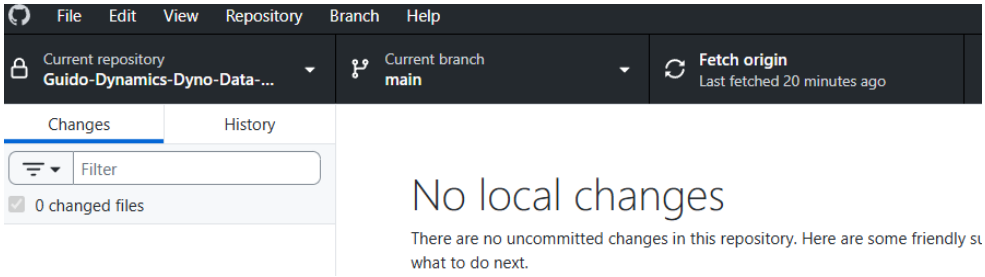
Select a path to your preferred location on your computer, and click "Clone".

[3] The default installation directory should be called "GrafanaLabs" in your Program Files directory

[4] If you are installing this after June 2026, the repository should be public, and should allow you to clone. If before June 2026, you may need to request access to the repository



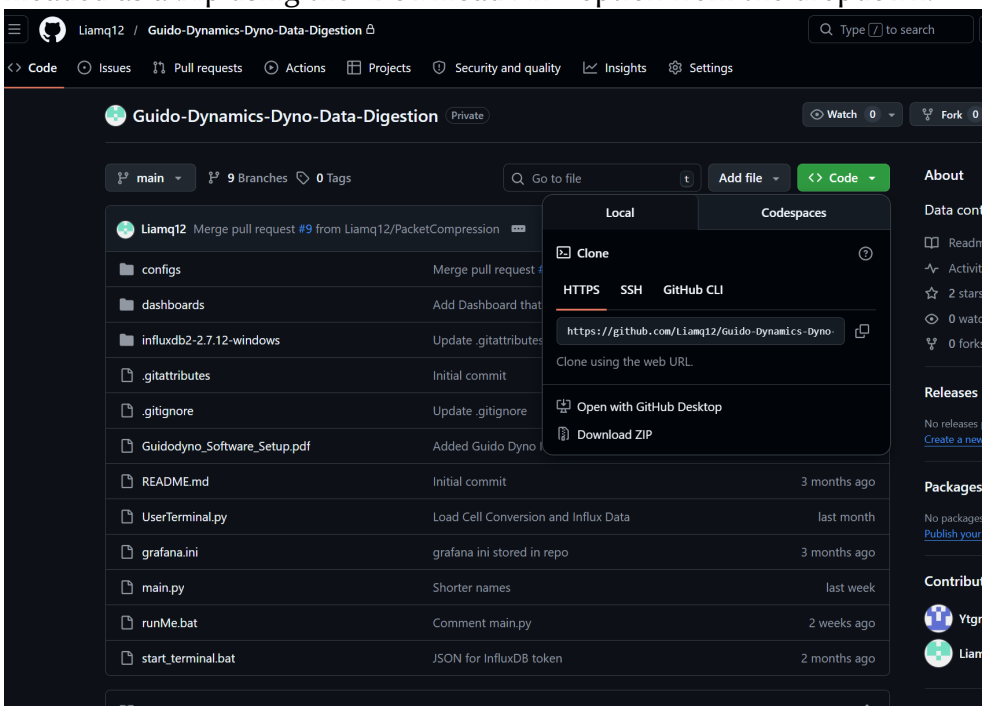
If prompted, click "Initialize Git LFS". Once the repository has cloned, click "Fetch Orgin" a the top of the Github desktop application. Make sure to fetch orgin periodically to get the latest software updates.



The Guidodyno software should now be installed in the directory you listed from the cloning step.

3.3.2 Online Repository

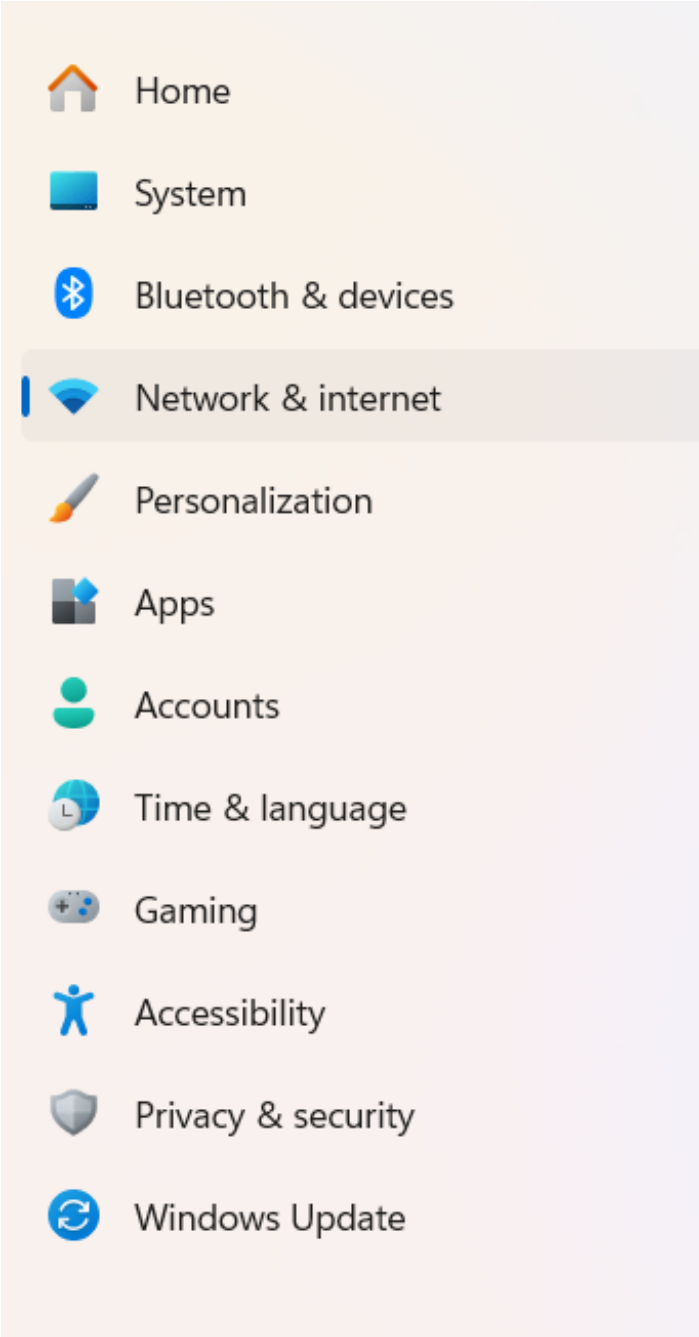
Navigate to 'https://github.com/Liamq12/Guido-Dynamics-Dyno-Data-Digestion'. From the green "Code" button, the user can see how the program can be downloaded using git if they are familiar. The program can also be downloaded as a .zip using the "Download ZIP" option from the dropdown.



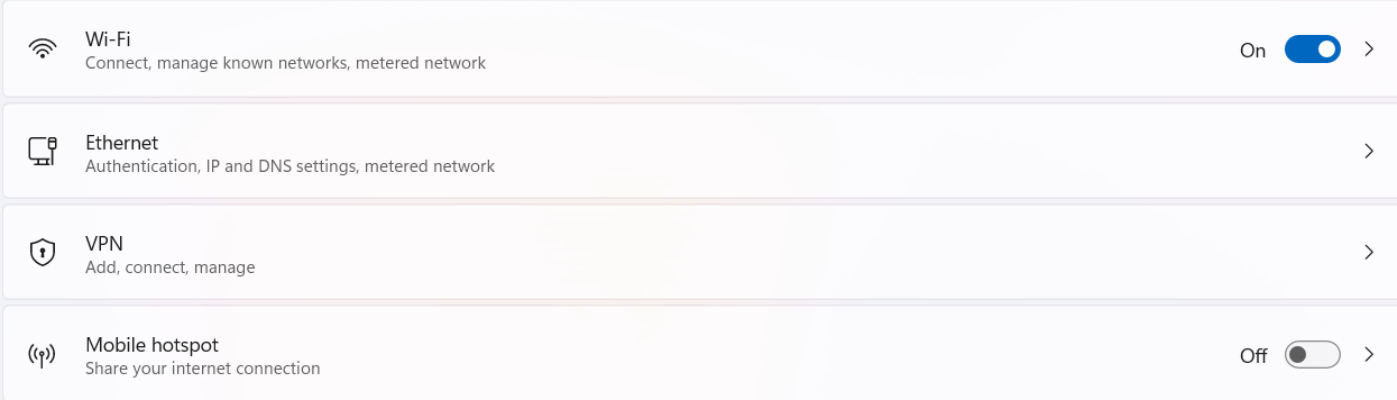
The program should be extracted, if applicable, and the folder location should be noted.

3.4 Network Settings

In the windows start menu, search for "Settings" and navigate to the "Network and Internet" tab.



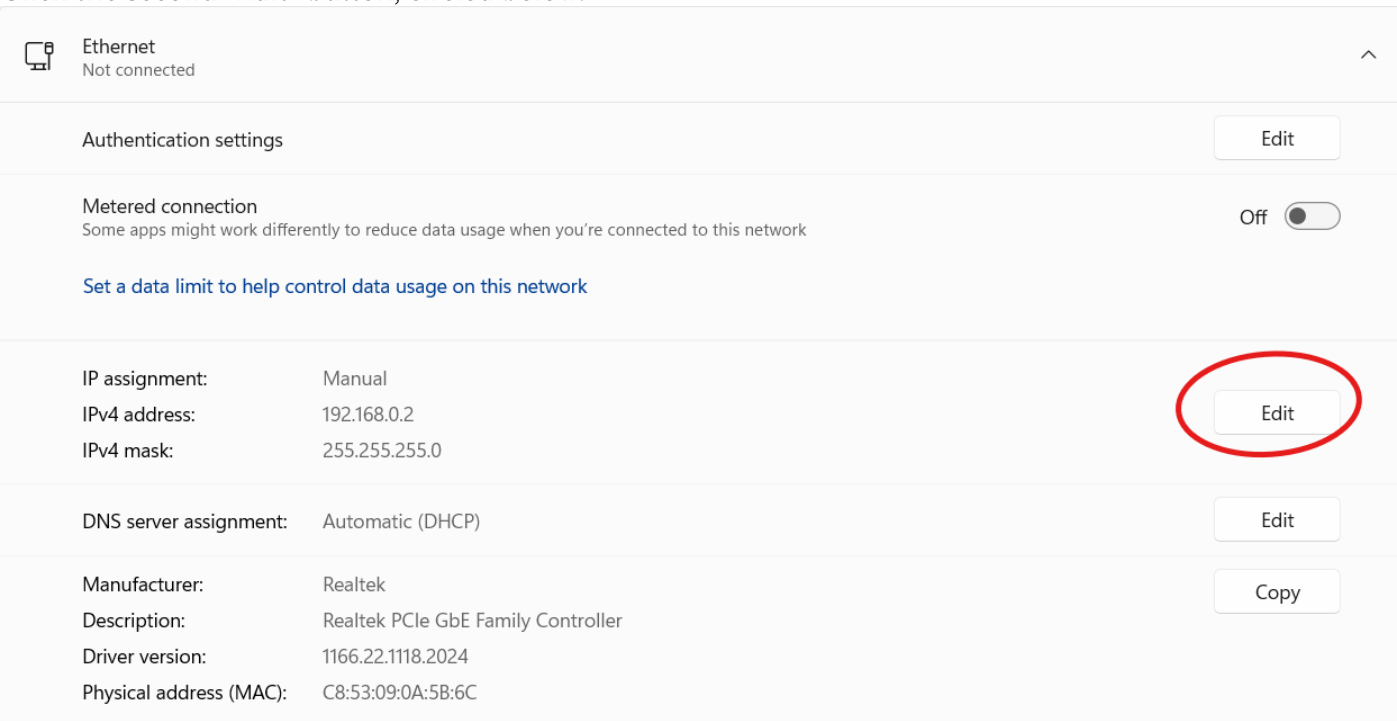
If you have a physical Ethernet port, click the menu button called "Ethernet"



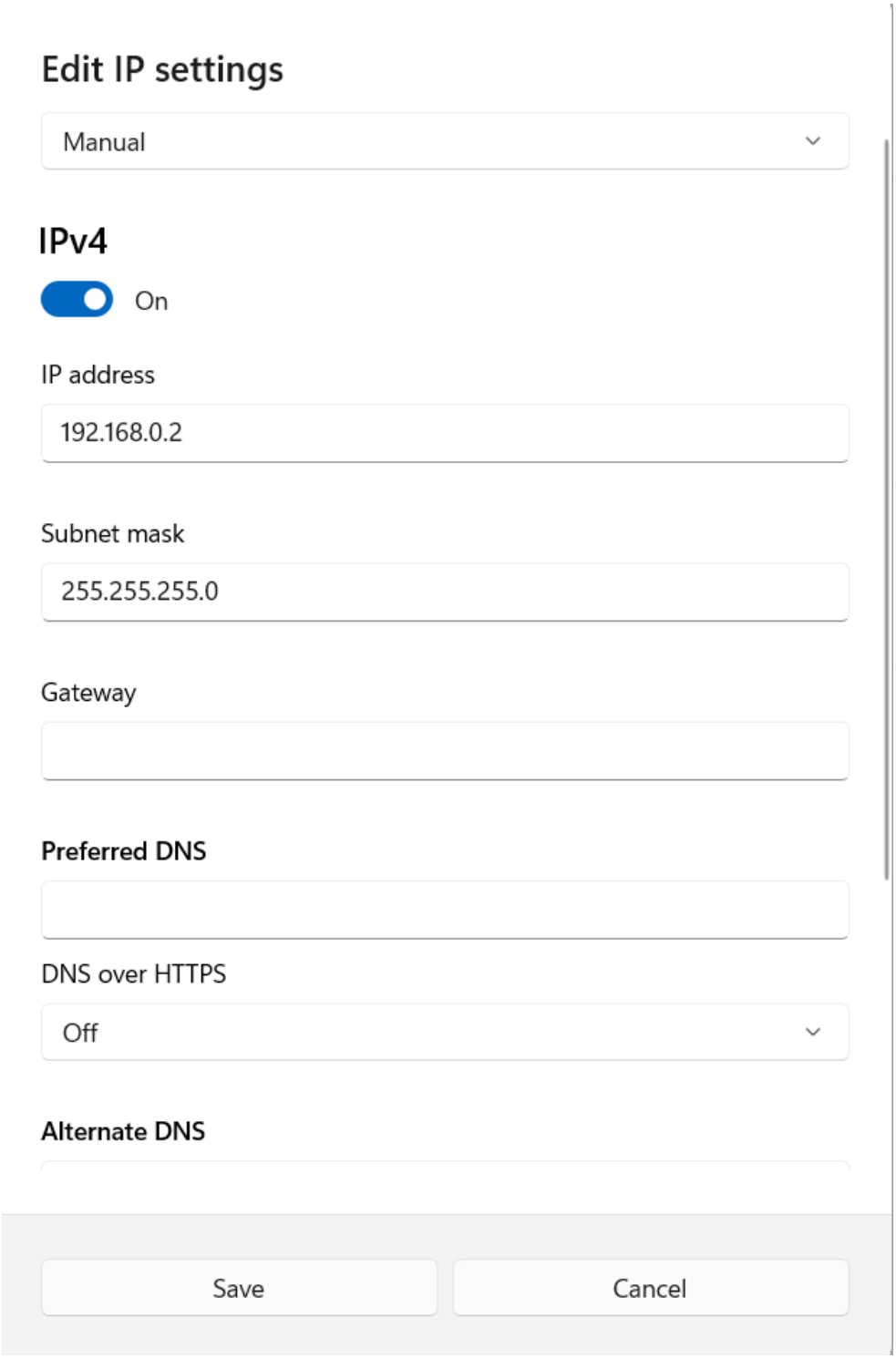
No Ethernet Menu

If you do not have an physical Ethernet port on your computer, you may need to navigate further menus to find this Ethernet network settings menu.
On Windows 11, the virtual Ethernet adapter can be found by clicking "Advanced Network Settings" at the bottom, and then clicking "Enable" on the Virtual Ethernet Adapter. IP address information is edited by clicking "View Additional Properties" within the adapter.
If you cannot find this menu, skip to step 3.5.

Click the second "Edit" button, circled below.



Under "Edit IP Settings", select manual.
Turn on IPv4.
For IP address, enter "192.168.0.2".
For subnet mask, enter "255.255.255.0".
Leave gateway field empty.
Leave perferred DNS field empty.
Under "DNS over HTTPS", select off.
Leave alternate DNS field empty.
Again, under "DNS over HTTPS", select off.
Turn off IPv6.
Click save.



3.5 runMe.bat

The runMe.bat file, located in the top level of your cloned repository, will start the necessary software to operate Guidodyno. Navigate to this file in file explorer, right click on the file, and run as administrator. [5] If a blue menu pops up, select "More Info", and then "Run Anyways".

This script starts the necessary software (InfluxDB, Grafana, UDPtoInflux Bridge "main.py", and the user terminal) and will spawn many windows. Leave this windows running while using the system.

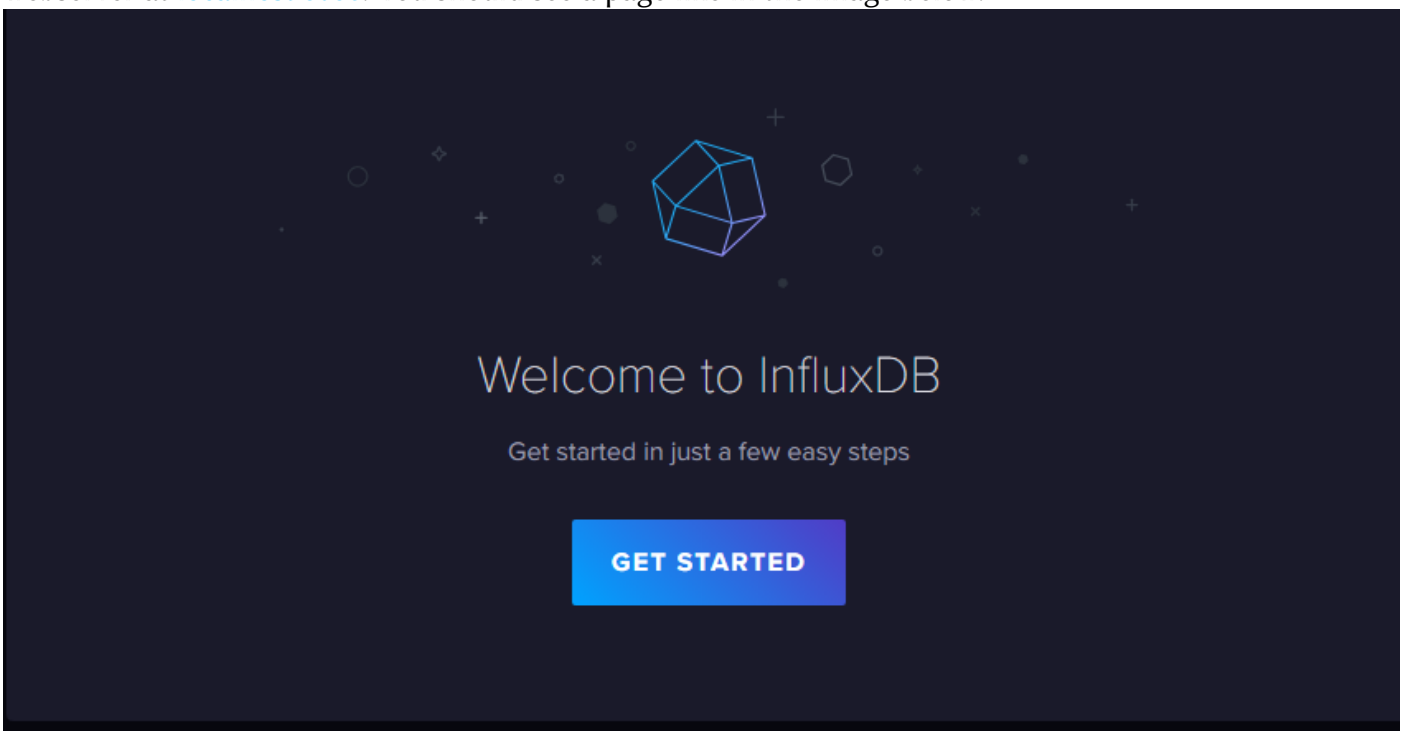
```
Starting UDP Ingest...
Starting User Terminal...
All processes started. Press any key to stop them...
Press any key to continue . . .
```

First Time runMe.bat

The first time running this script often requires "kicking" (press enter) the terminal. If the window is stuck at "The Grafana service was stopped successfully", press the enter key once to restart the program. Additionally, press "Allow" for any prompts that pop up.

3.6 Configuring InfluxDB

With the script running from the previous step, open a new browser tab and go to the InfluxDB local webserver at localhost:8086. You should see a page like in the image below.

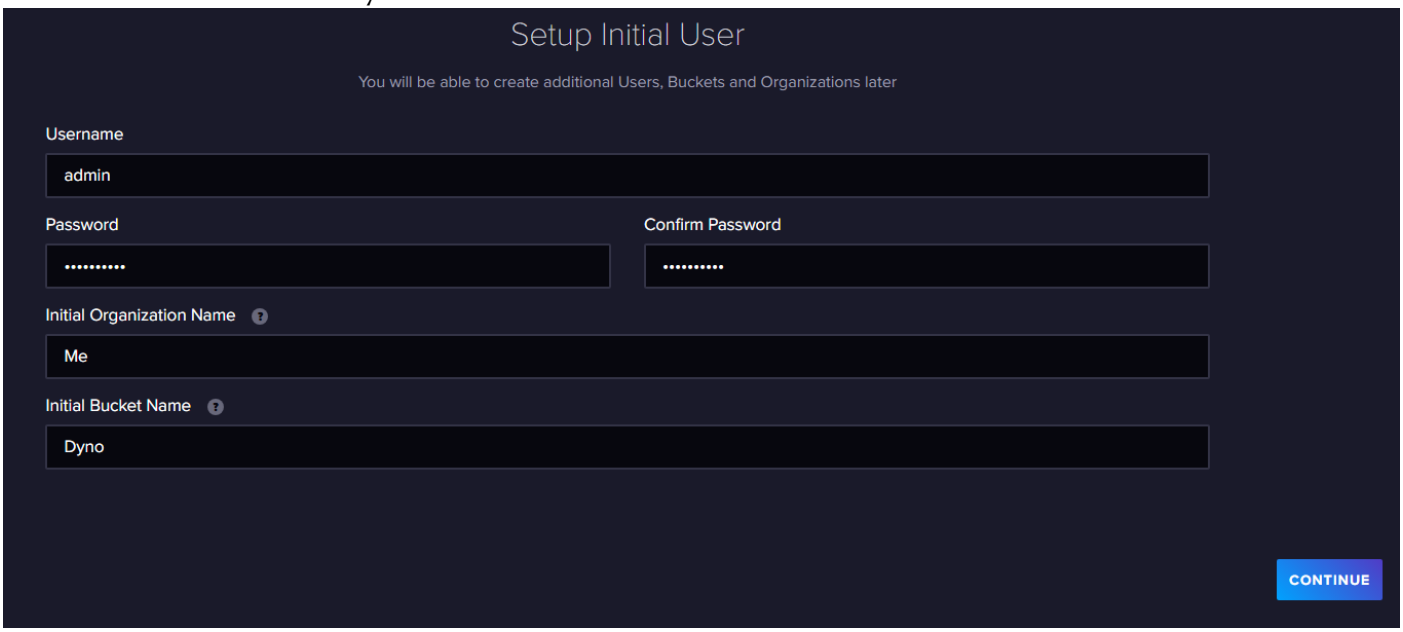


Error: Localhost Refused to Connect

Ensure the runMe.bat program is running (as administrator), and that you have fetched the latest version of the repository from Github Desktop.

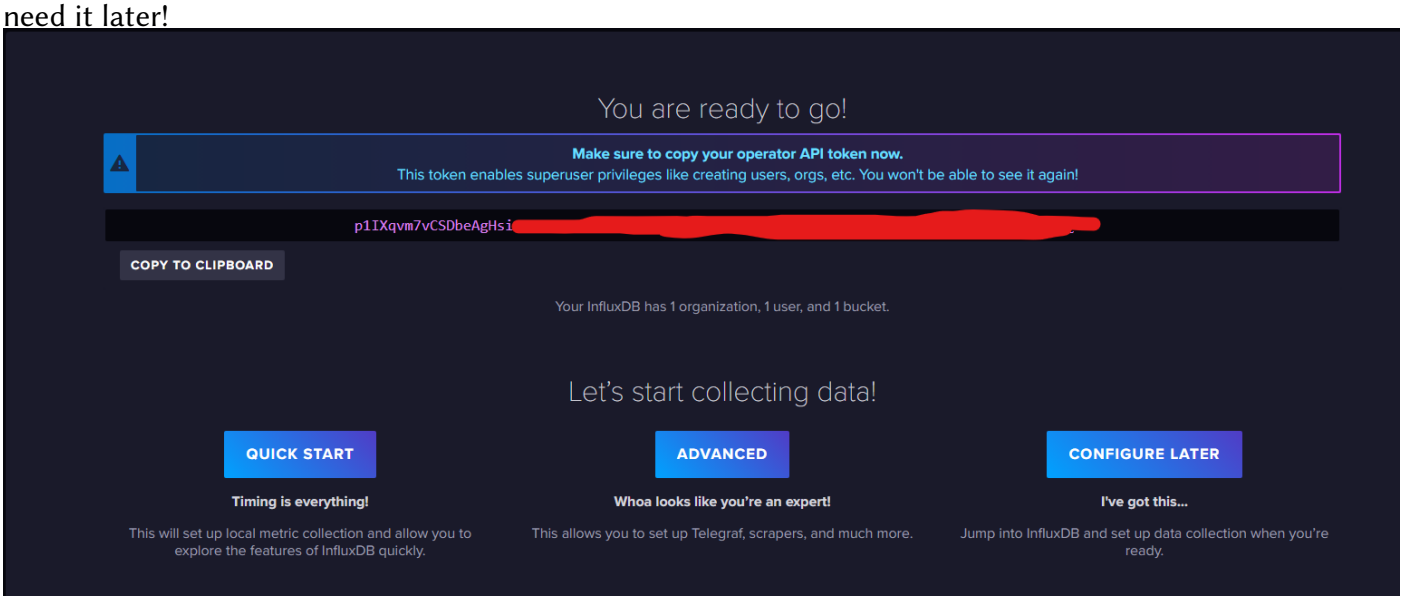
[5] This file is located with your other files cloned from GitHub. Unless changed in the clone step, the default location is: `C:\Users\[YOUR_USERNAME]\Documents\GitHub`

Click "Get Started" and make a username and password. [6] Set "Initial Organization Name" to "Me", and "Initial Bucket Name" to "Dyno". Click "Continue".



Since this information is stored locally, it is highly recommended to allow your browser to save your username and password.

You will now see your unique superuser token. This is only shown once, and you will not be able to view it again after moving from this page. Copy the token and save it somewhere you will remember, you will need it later!



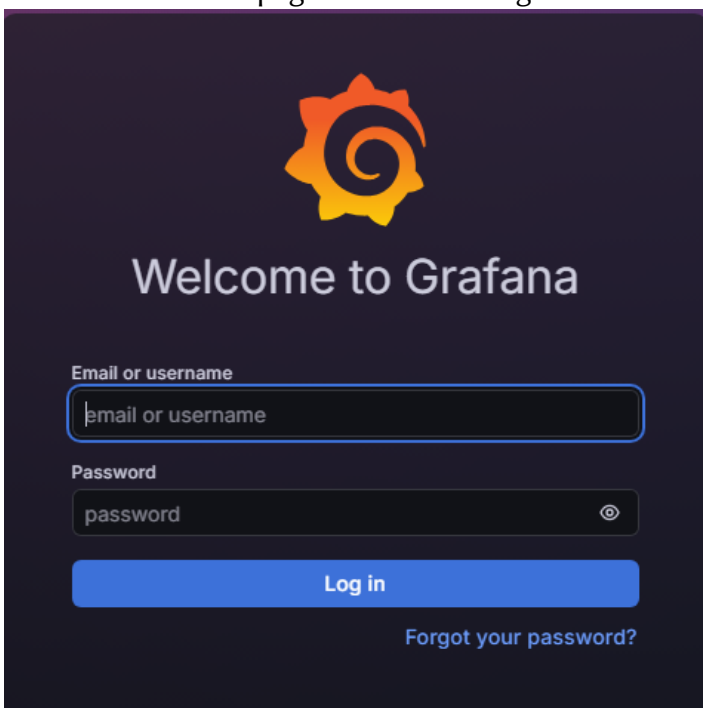
Click "Configure Later" once you have saved your token.

You are now done setting up InfluxDB. Feel free to close this tab.

[6] User data is stored locally, and not in the cloud or the Guidodyno Data Digestion repository. There is no way to recover a lost username/password, so make it simple. It is only accessible when signed into your Windows user account.

3.7 Configuring Grafana

With the runMe.bat script running from step 3.2.1, open a new browser tab and go to the Grafana local webserver at localhost:3000. You should see a page like in the image below.



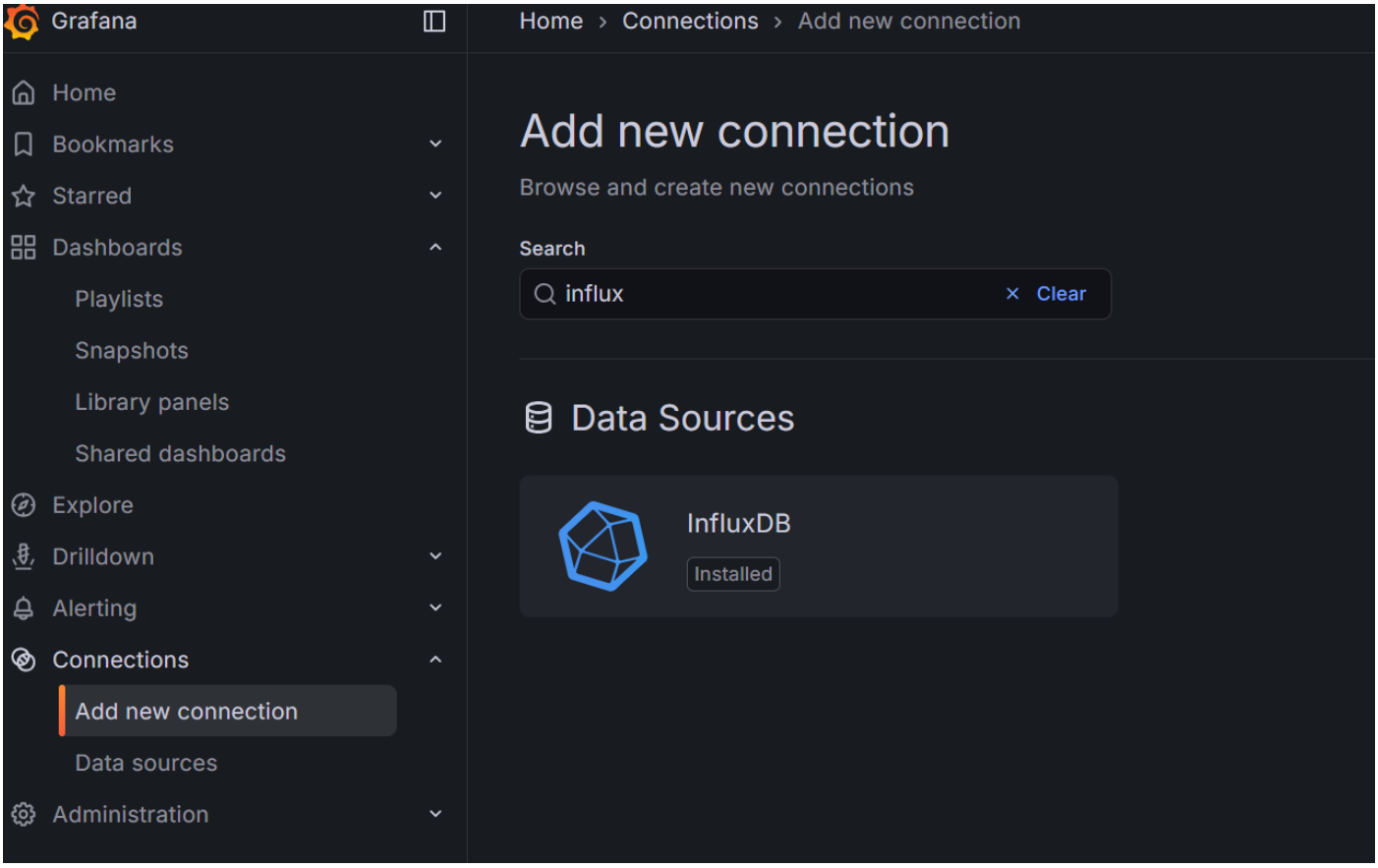
Error: Localhost Refused to Connect

Ensure the runMe.bat program is running (as administrator), and that you have fetched the latest version of the repository (With Git LFS enabled) from Github Desktop.

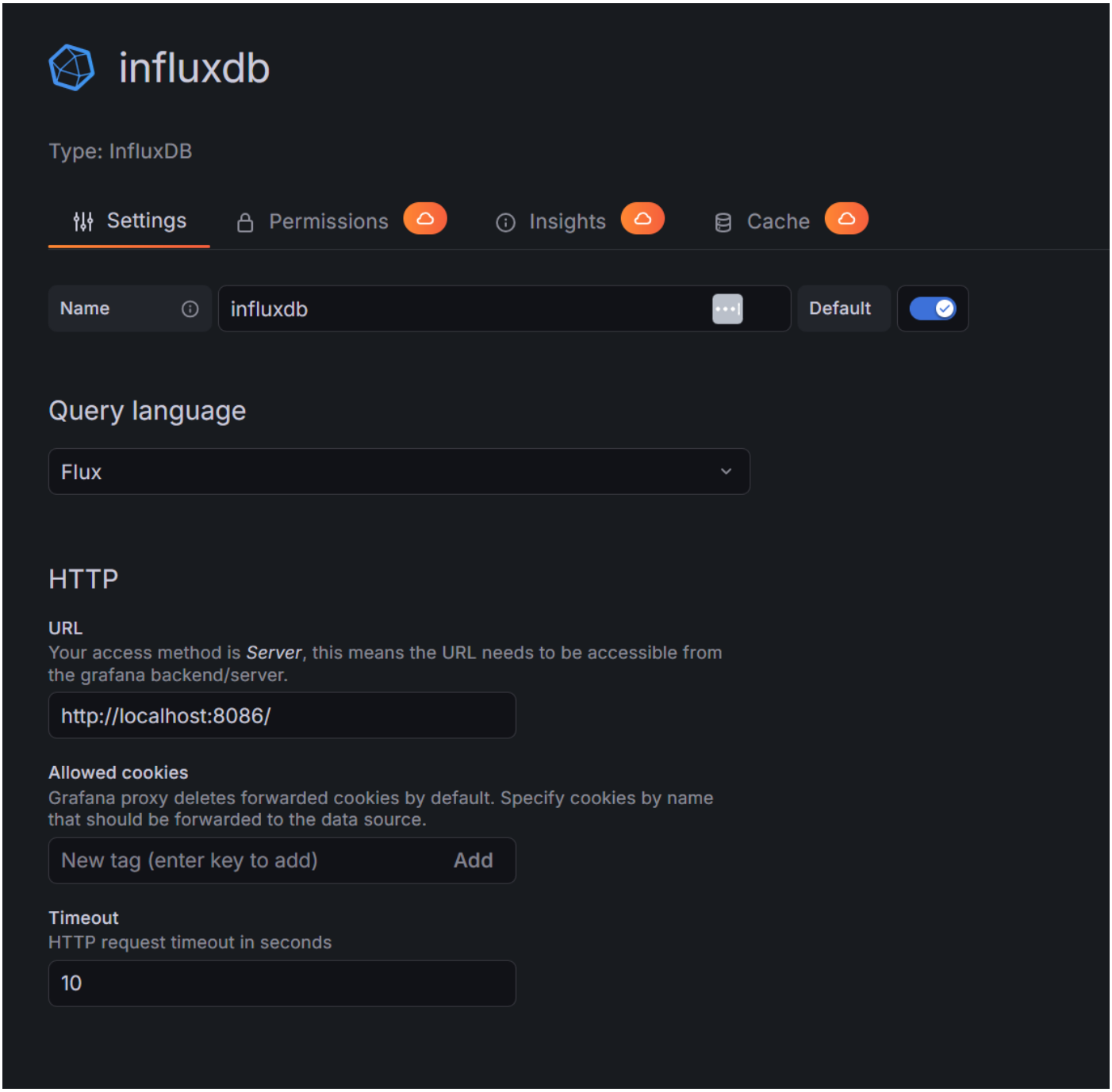
The default username is "admin", and the default password is "admin". It will then prompt you to change the password. Like InfluxDB, this is only accessible within your Windows user account, so retype "admin" as your new password.

3.7.1 Adding InfluxDB Data Source

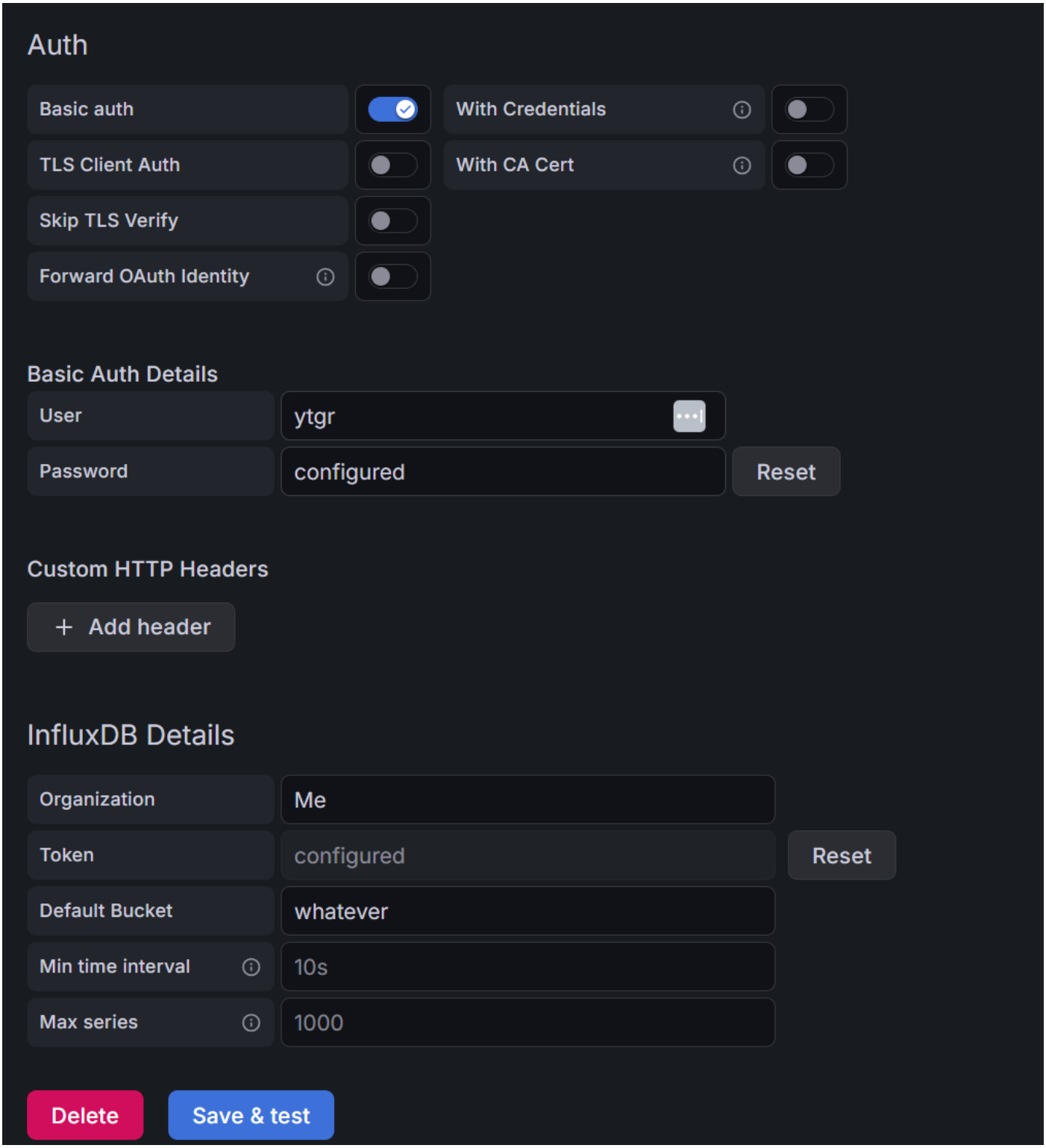
In Grafana, on the left side menu, navigate to Connections->Add new connection. In the search by, type in "influx". Choose InfluxDB, and click "Add new data source".



Configure the InfluxDB source as below, with the Query Language being Flux, and the URL being "http://localhost:8086/. Ensure the default setting is enabled.



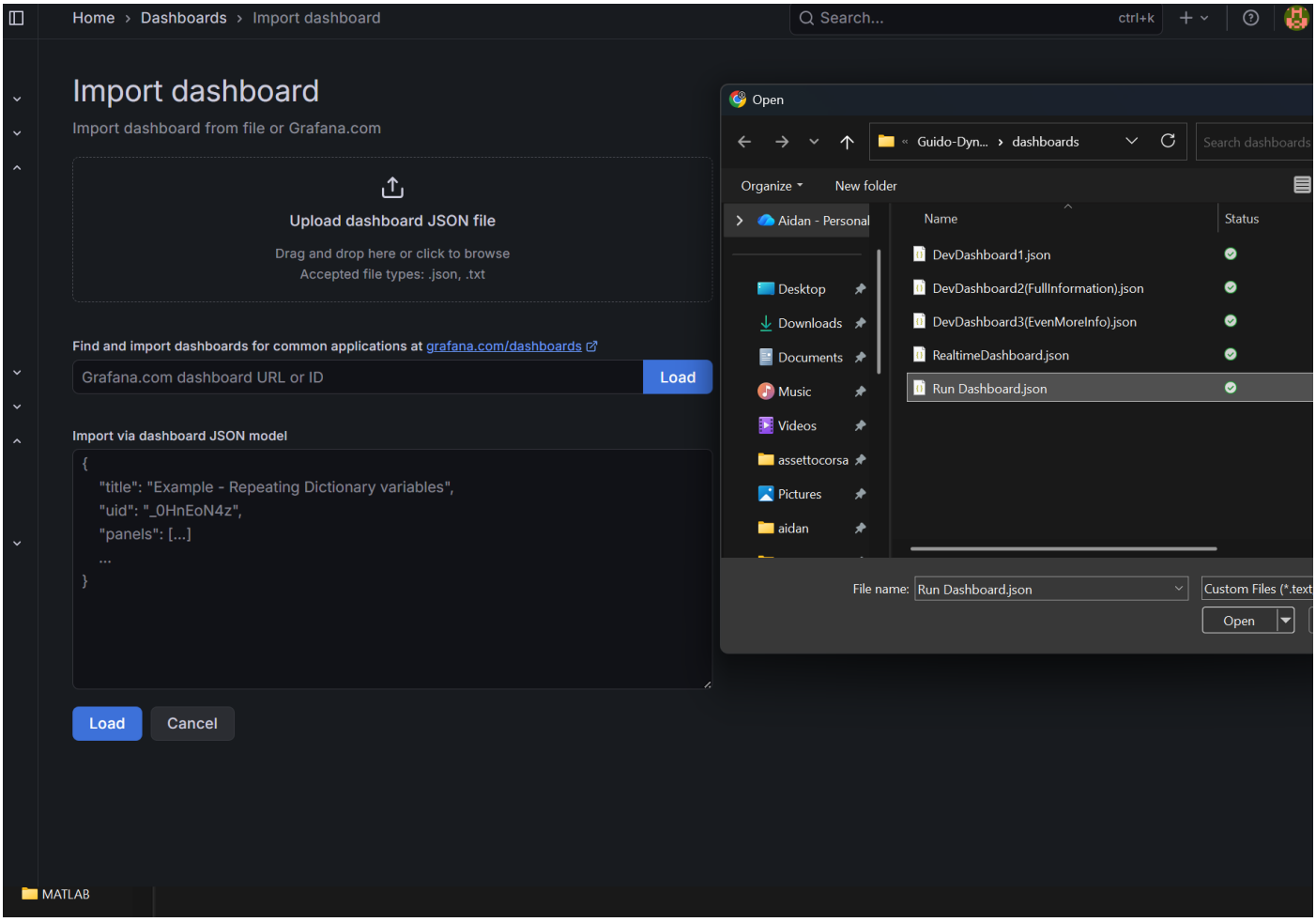
The bottom half of the InlfuxDB setup is shown below. Ensure the Basic Auth switch is selected. Enter your InfluxDB username and password in the appropriate fields. Enter the Organization, Token, and Bucket in the appropriate fields as well. Then select save and Test. Grafana should now be setup to pull data from InfluxDB.



3.7.2 Loading Dashboards

Within the downloaded program folder, there is a "dashboards" folder. Here lie the "RunDashboard.json" and "RealtimeDashboard.json", which can be imported into grafana. These are already setup for you, where the run dashboard groups data from individual pulls, and the realtime dashboard provides realtime data for debugging and other use cases.

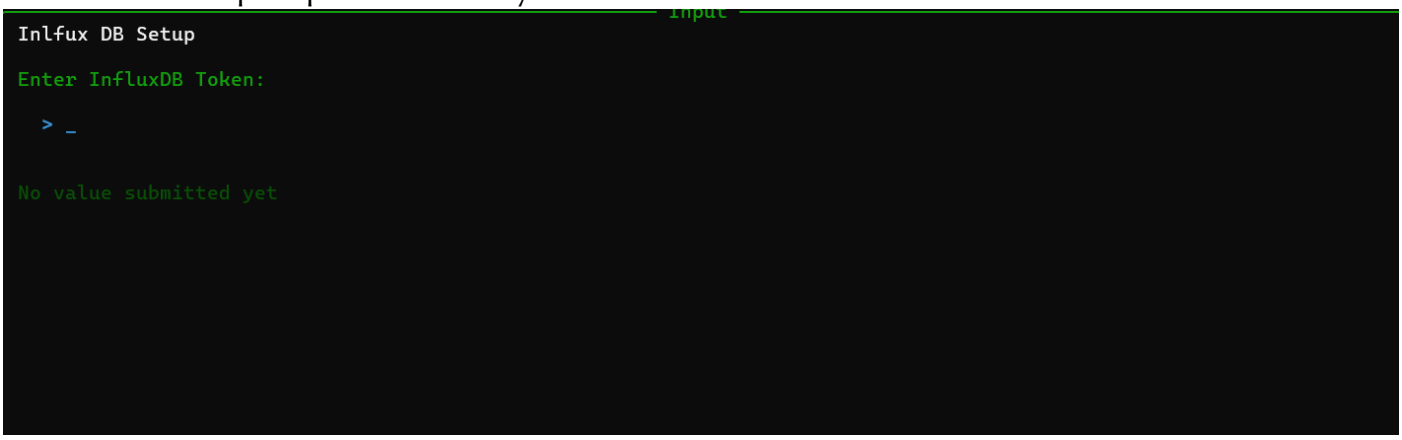
In the Dashboards tab of Grafana, select New->Import. From here, import the dashboards you want. For this tutorial, load the Realtime dashboard.



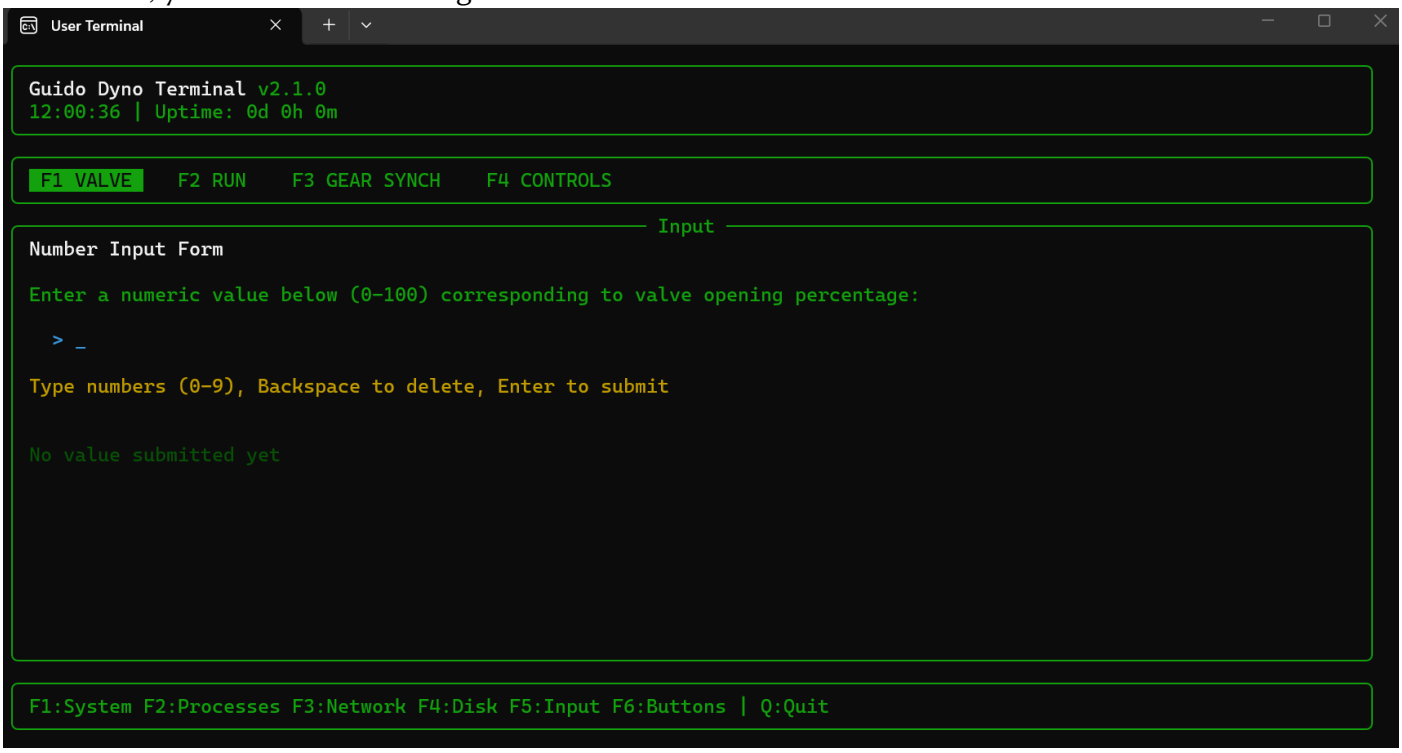
Within these dashboards, there is a textbox at the top titled "Bucket Name". Enter the bucket name you set up in the influxdb setup so that the dashboards can properly pull from your data.

3.7.3 Testing Grafana

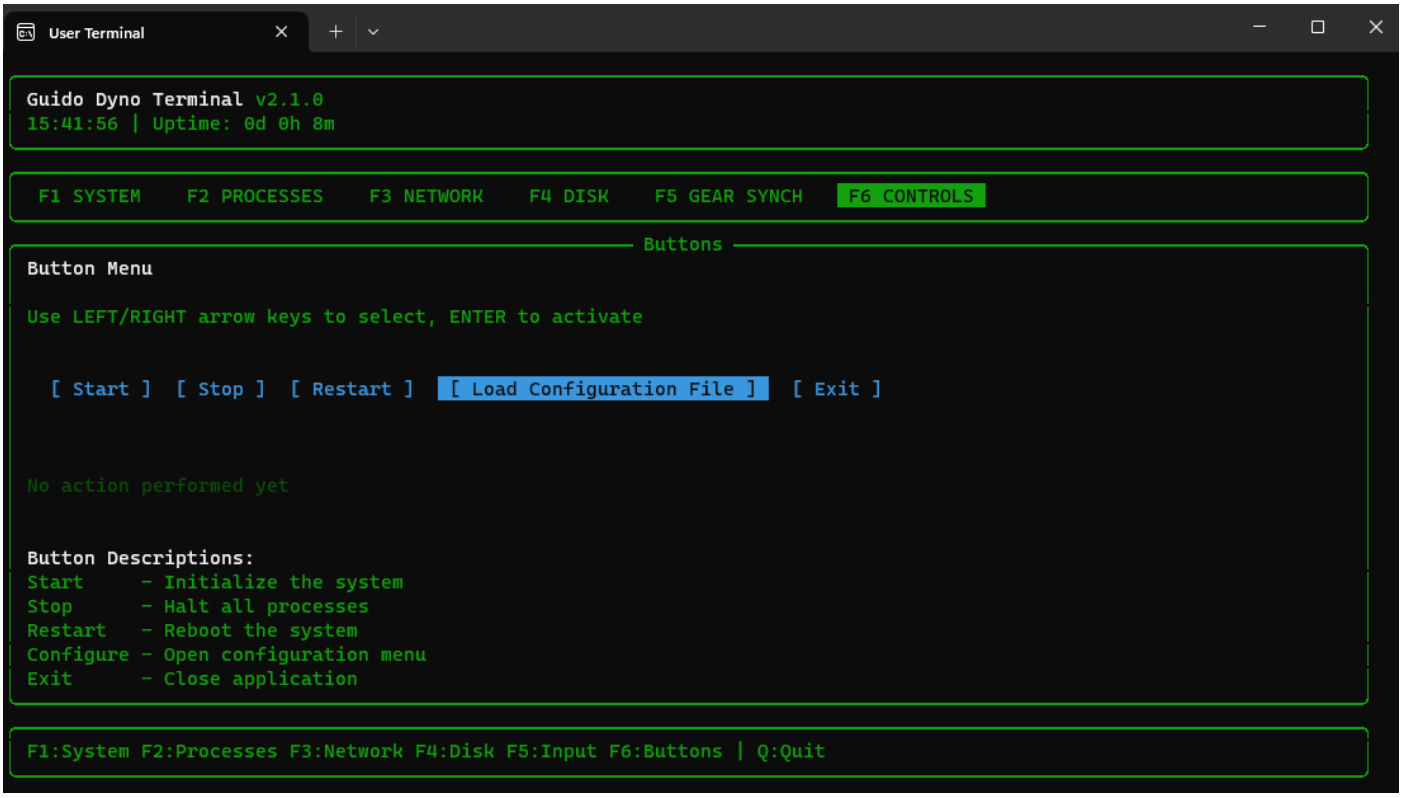
Run the runMe.bat program in the software folder as administrator, and look for the user terminal running in a CMD window. The splash screen looks like the image below. Enter your InfluxDB token, organization, and bucket when prompted. This is only done once.



Afterwards, you should see an image like below:



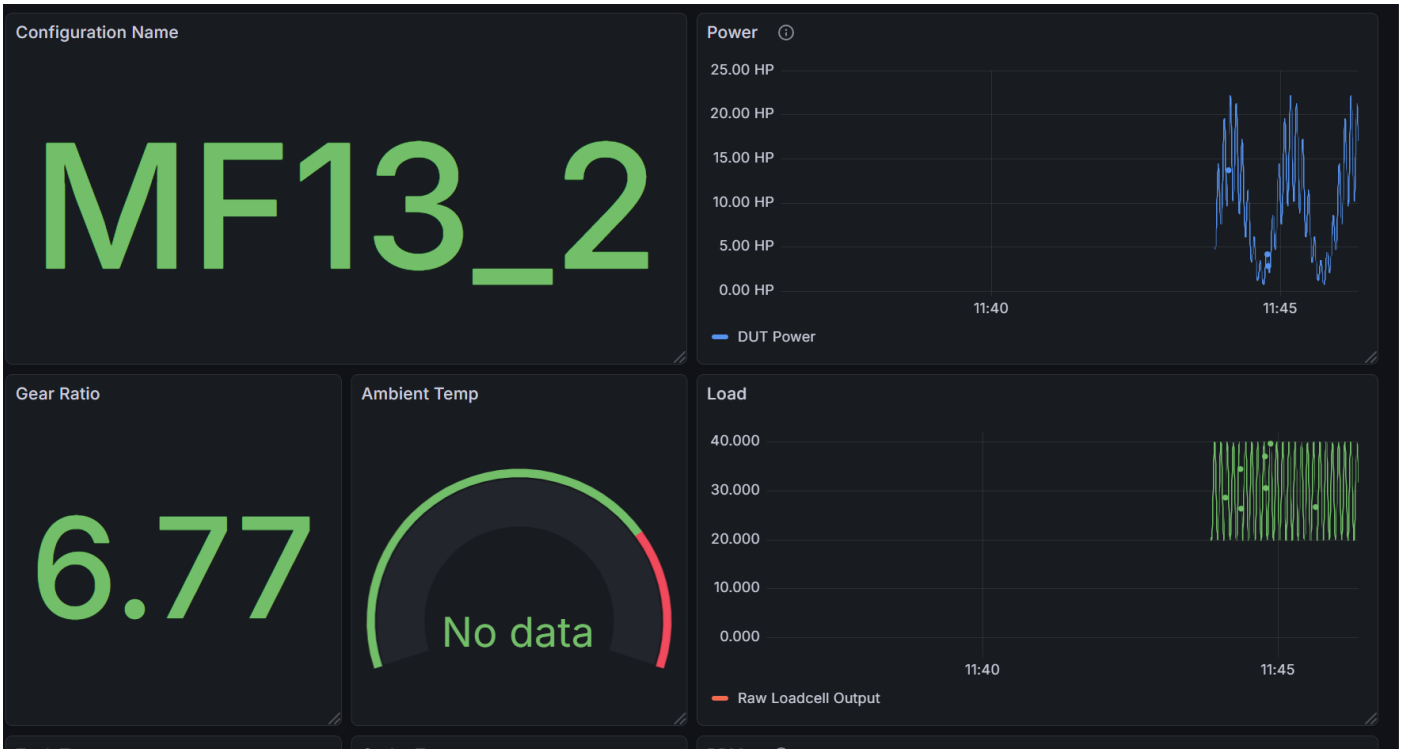
Click F6 to move to the Controls tab, then use the arrow keys to move to "Load Configuration File". Press enter to select.



Navigate to the config folder in the repository folder and open "VehicleConfiguration.json". This config folder is where Dyno configuration files are stored.

Back on the Grafana dashboard, click refresh in the top right[7]. Your stat visualization should now display some data! Once a configuration file is loaded, the configuration name should be uploaded. If the computer is not connected via Ethernet to the dyno, fake data will populate within the dashboard to ensure all the internal systems are working:

[7] The refresh button in the Grafana window, not the browser refresh button



If that worked, your system is configured and ready to collect data from the Guidodyno!

4 Software Use

4.1 Starting the System

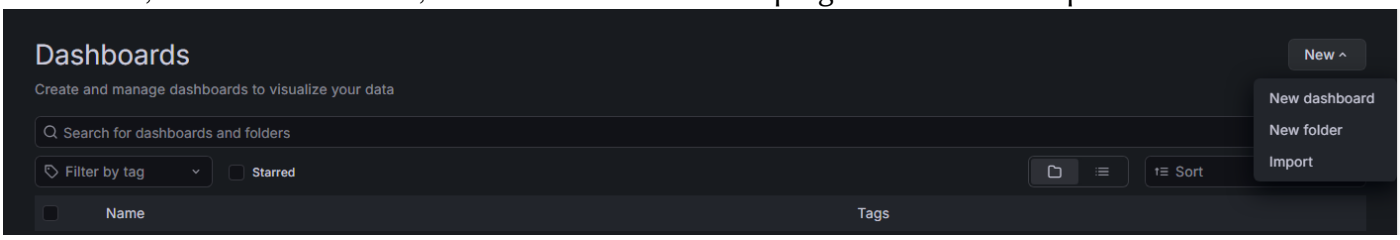
The runMe.bat script, located in the top level repository folder, starts all necessary software to run the Guidodyno. Make sure the dyno is connected to your computer over Ethernet if you are collecting data before running this script. Always run the script as an administrator.

Ensure the pre-operative checklist in section 2.4 has been executed before operating the dynamometer.

Thats it! Go to localhost:3000 to view live data. Go to localhost:8086 to download and view 5m+ data.

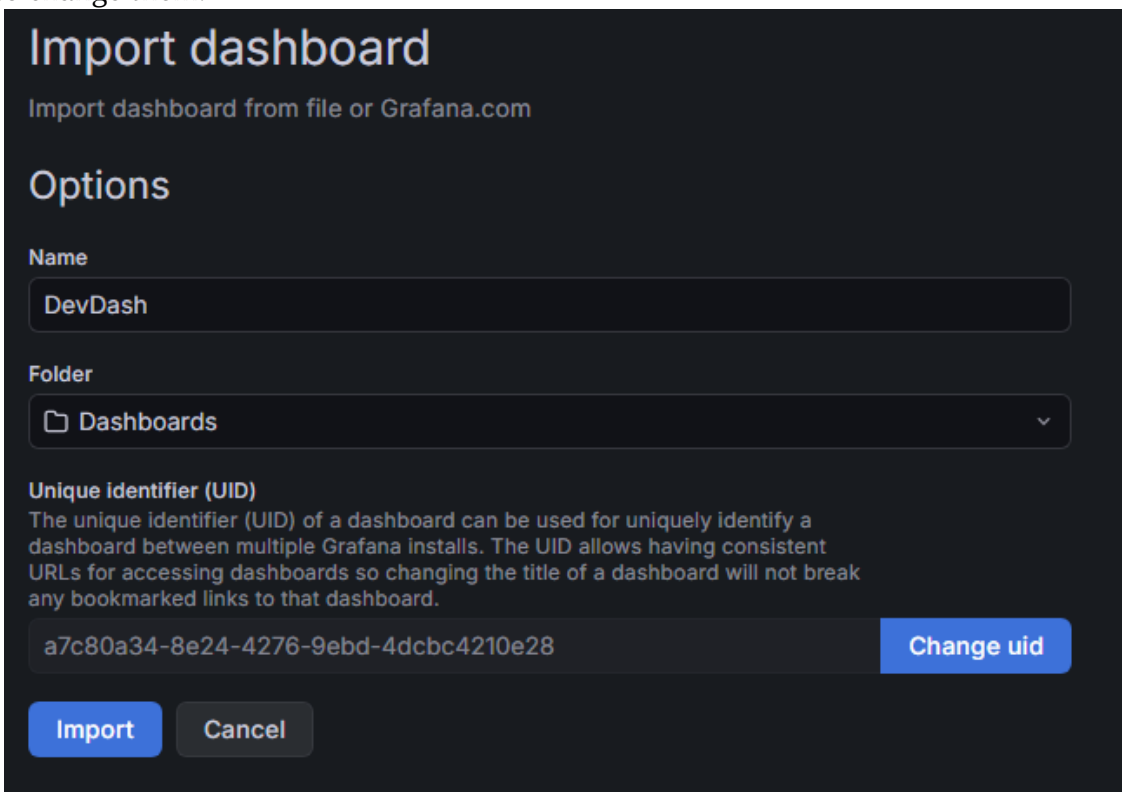
4.2 Importing Grafana Dashboards

In Grafana, select "Dashboards", and select "New" in the top right. Then click "Import".



Click "Upload dashboard JSON file", and navigate to the Dyno software repository. There is a top level folder called "dashboards" that contains dashboards made by other Dyno users you can use. "RealtimeDashboard.json" is one that is used for testing the system.

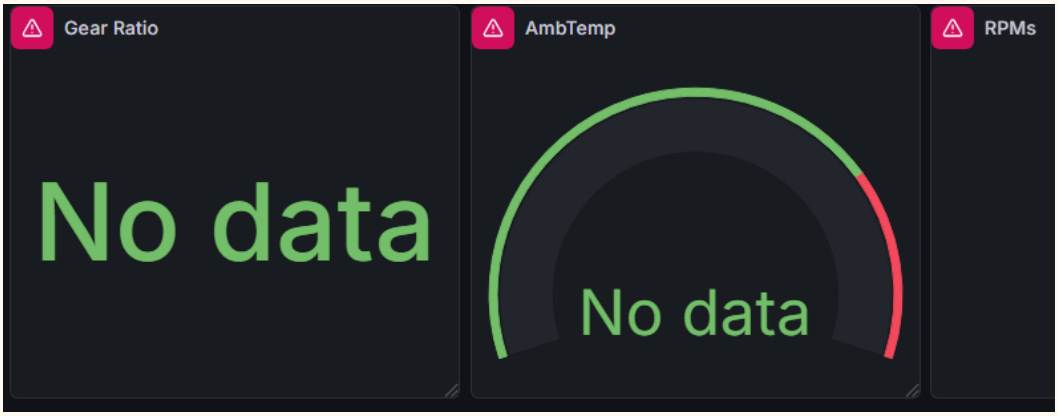
Select the dashboard you want to import, and give it a name. Leave the Folder and UID fields alone unless you need to change them.



Click "Import". The dashboard will now load.

Make sure to update the bucket name variable at the top with the bucket name that you used in InfluxDB setup.

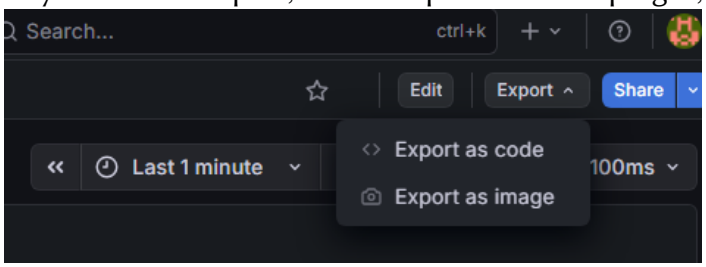
Bucket Name Mismatch



Some Dyno users use different bucket names for various reasons. A likely cause of visualization errors (red triangle in top left of visuals) is the wrong bucket name. When using either the Run Dashboard or Realtime Dashboard, the bucket name can be changed using the global variable at the top. When using other dashboards, click the three dots in the top right of a visualization, click "Edit", then change the bucket field to your bucket name. This needs to be done for each visualization, but only once after importing a dashboard. It is now saved on your computer, and can be exported with your bucket names.

4.3 Exporting Grafana Dashboards

With the Grafana dashboard you want to export, select "Export" in the top right, and click "Export as code".



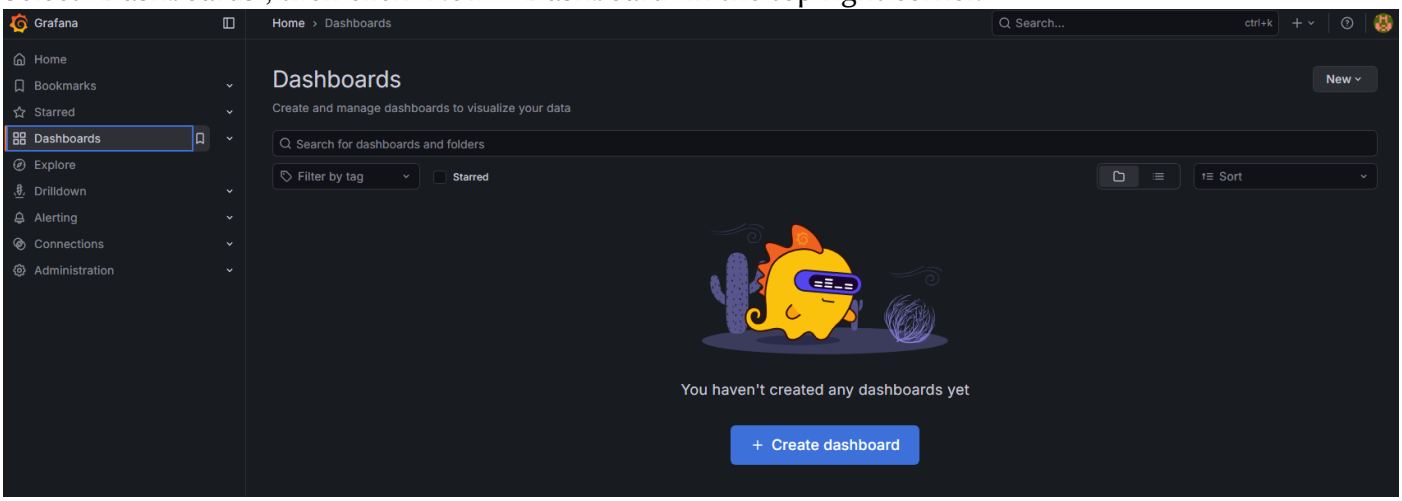
Click "Download file" in the bottom left of the download menu.

There is an shared dashboards folder in the software repository you can add your dashboard to if you would like to share with others.[8]

[8] Folder is named "dashboards"

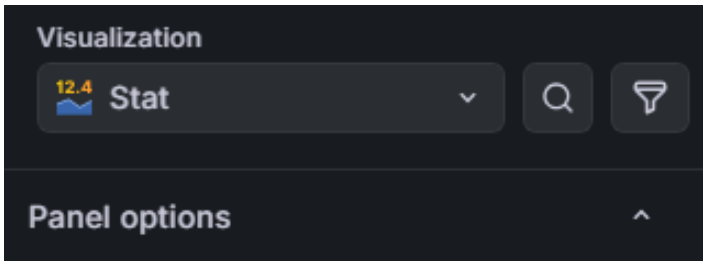
4.4 Generating New Dashboards

Select "Dashboards", then click "New">"Dashboard" in the top right corner.



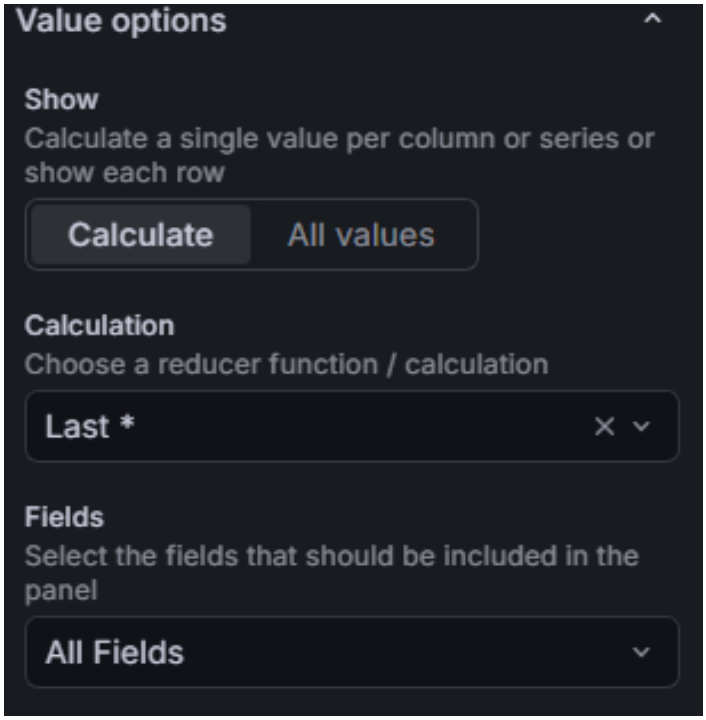
Click "Add visualization", then click on InfluxDB [9]

Select "Dashboards" from the left menu, click "Create dashboard", click "Add visualization", then click "influxdb" from the "Select data source" menu. This will open a new visualization pulling data from Influx.

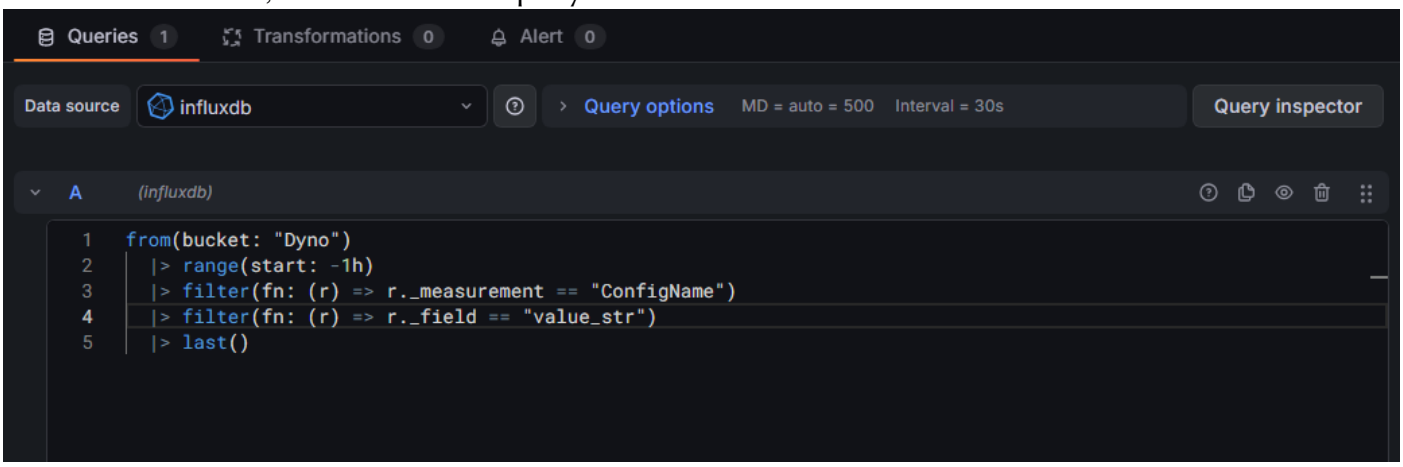


[9] Imported from step 3.7.1.

Also in the right side menu, under "Fields", select "All Fields"



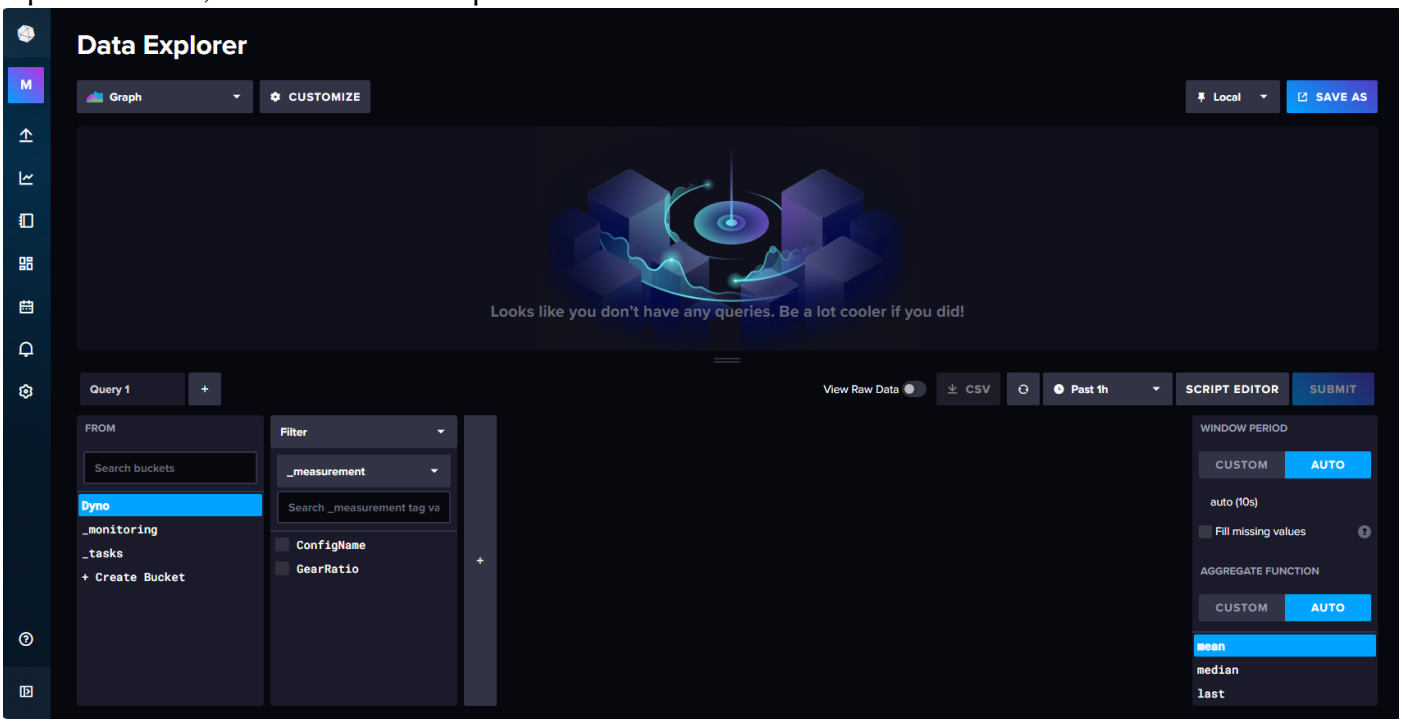
In the main window, an InfluxDB flux query can be written to retrieve data from InfluxDB.



(More info on this TODO)

4.5 Exporting Data to CSV

Open InfluxDB, and select "Data Explorer" form the left side menu.



Using the check boxes, select the data you want to export. Once selected, select how far you want to look back, set your window period (data rate), and click the blue "Submit" button. Once complete, click the CSV download button, next to the "View Raw Data" slider. This will download your selected data with UTC time basis information.

4.6 View Previous Data with Grafana

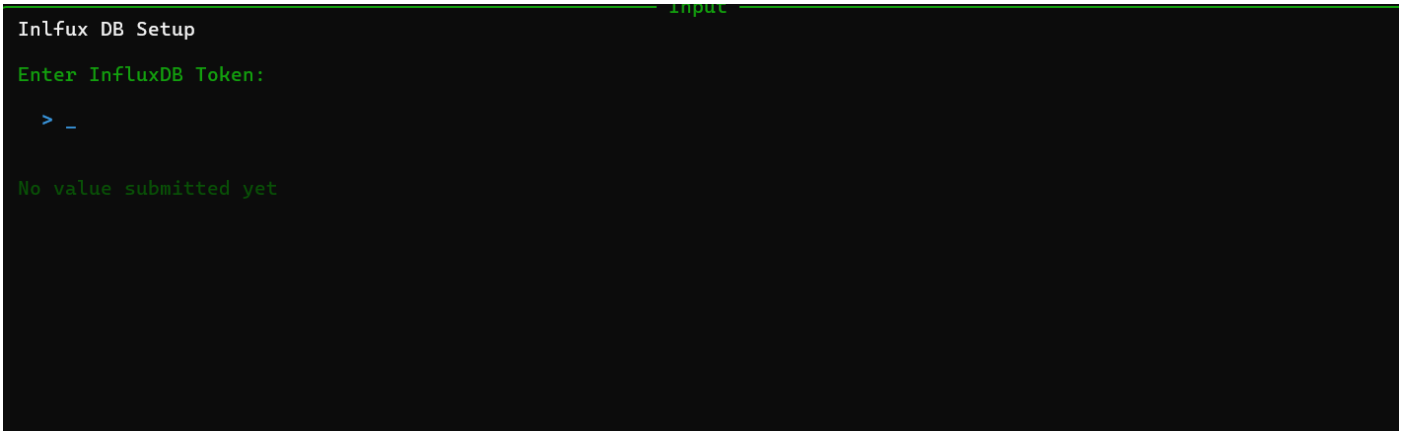
TODO

4.7 User Terminal

The user terminal application allows engineers to manually control the dyno, and load automated ramp sequences.

4.7.1 Setup

The first time the user terminal is opened, the user must paste their InfluxDB token, bucket name, and org name into the terminal. This is a one time setup step.



Paste your InfluxDB token, bucket name, and org name from step 2.6.

This data can be changed anytime by going to F4>Setup InfluxDB, or by editing the config file configs>system>influxdb.txt.

The user terminal must be programmed with the stepper controller's pulses per revolution setting, as set with the DIP switches on the controller. This is set to 6400 by default, and the user terminal initializes with this value. **Only change this value F4>[Setup Valve] if it is changed on the stepper controller.**

Stepper Controller

The stepper controller is located inside the central control panel. Service must be performed only by qualified personnel with demonstrated competence in working with mains electrical systems.

4.7.2 Gear Ratio Calculation

Load the "GearRatioSyncConfig.json" file provided. With a system on the dyno (engine or vehicle), hold the throttle down and wait for the RPM to stabilize. Enter this RPM into page F2 while the RPM is held. Record the calculated gear ratio. In the vehicle configuration file, enter this calculated gear ratio in the "Gear Ratio" field of the JSON.

JSON Formatting

The gear ratio must be inputted as a numerical value, ie. there must not be quotations around the number in the JSON file.

4.7.3 Manual Load Control

The user terminal can manually control the dyno load by changing the position of the valve. To do this, go to window F1, and type a value 0%-100%. This will set the valve to this position until commanded otherwise.[10]

4.7.4 System Configuration

To load a system configuration file, first go to the configs folder, and create a vehicle file. VehicleConfiguration.json is provided as a template you can use to start.[11]

Name this file, and change the value for the field "Gear Ratio". See section 3.6.2 on how to calculate the gear ratio if it is unknown.

Gear Ratios

The gear ratio is affected by any of the applicable factors: gear selection, final drive, tire size, roller size, sprocket size, and more. As such, the user cannot simply "plug in" the selected gear ratio from a spec sheet, and it is recommended to undergo the gear ratio calibration outlined in the above section.

If your system has multiple gears, create a file for each gear with the relevant gear ratio.

The "Name" and "Version" fields are intended for the user to be able to differentiate between runs. It is recommended to keep the "Name" field abbreviated for readability and include the gear ratio. For example, the name "MF13_G3" may denote the Mines Formula 13th chassis in 3rd gear.

The "ID" field is used by the database, and must be unique for each vehicle and configuration, although these can be arbitrary.

[10] This will not work if the dyno is in PID mode or ramp mode.
[11] Only use the top section of this config file. The bottom is for RPM ramp settings.

Enter the desired run plan in the bottom section of the vehicle configuration. See the "HoldEx" and "RampEx" files in the Run Plans folder. The information from these files can be copied and pasted into the vehicle configuration. Alternatively, the run plan section of the vehicle configuration file can be deleted, and the user will need to load run plans separately. Be cautious of commas in the JSON file.

Before operation, go to F4>Load Configuration File and select your configuration file.

4.7.5 Run Plans

To load a run plan, first create an run plan configuration file. The files RampEx.json and HoldEx.json, located in the Run Plans folder, contain sample run plan configurations for a ramp and RPM hold, respectively.
These run plans can be loaded in the F4>Load Run Plan menu. Run plans loaded in this manner take precedent over any run plan integrated in a configuration file, if applicable. If the configuration file does not contain a run plan, it must be loaded manually as described in this section.

RPM Values and Gear Ratio

Run plans, integrated into the configuration file or loaded independently, use the engine RPM for control. This means the loaded gear ratio must be correct for the engine RPM desired.

5 Further Information